Numerical Investigation on Electrical Contact Resistance of NiPdAu-Plated CPU Package Pad and NiAu-Plated Second-Level Interconnect Socket Contact Interface

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***Abstract*—*High performance electronic package requires good control of electrical contact resistance at the package pad and second level interconnect (SLI) socket contact interface for power delivery and thermal management. In this paper, a modeling framework incorporating electrical contact theories is developed to investigate contact resistance of the given multi-layer plating system. The effects of some potential electrical contact resistance variation contributing factors are studied and quantified. The parameters studied include contact normal force, microhardness and thickness of constituent plating layers, phosphorus content percentage in electroless plated Ni layer, metallic nanofilm size effect, and existence of NiO contaminant film.*

***Keywords*—** *Electronic package pad, socket contact, electrical contact resistance, multi-layer plating, modeling*

I. INTRODUCTION

In the semiconductor industry, NiPdAu surface finish has been widely used for printed circuit board (PCB) and electronic package pads due to its superior solderability and reliability. Gold is unlikely to interact with environmental elements to form an electrical insulating film, and therefore it is resistant to corrosion. Nickel and palladium are utilized as under plating layers to prevent the base copper from diffusing into the surface finish. In a sliding type of socket (e.g., Land Grid Array socket having contact interface wipe) and connector application, the gold at interface can provide stable and low electrical contact resistance.

The demand for increasing system performance of CPU package places challenges on power delivery and thermal management. As a result, it is important to predict and control electrical contact resistance at CPU package pad and SLI socket contact interface in order to meet system requirements. Specifically, the surface finish technology studied in this paper is Electroless Nickel and Palladium Immersion Gold (ENIPIG) for CPU package pad and Electroplated Nickel and Gold for SLI socket pin. The main difficulty of managing the electrical contact resistance at this interface is its complex attributes from many areas: plating process and composition, plating layer thickness configuration, hardness and roughness, system mechanical design impact on contact normal force, thermal and chemical history of the parts during the manufacturing process, as well as plating wear due to cyclic wiping, temperature and environmental exposure in the field. Many numerical and experimental methods have been employed to investigate electrical contact resistance and electrical properties of thin film materials, focusing on individual or a few aspects of these factors. The well-established Holm electrical contact theory [1] showed how to calculate constriction resistance of bulk to bulk half space interface. Williams and Greenwood developed a plating factor theory to study constriction resistance of single layer plating interface [2]. Other factors such as surface roughness, electrical properties of thin metallic films, plating process and properties have been investigated in [3-9].

However, what has been lacking is integration of a modeling framework with comprehensive theories for multi-layer plating system. In this study, a modeling framework incorporating theories of abovementioned factors is developed to predict electrical contact resistance of the given multi-layer plating system. Impacts of several hypotheses are investigated and quantified, including: contact normal force, plating material microhardness, electroless Nickel plating layer composition, metallic nanofilm size effect, Au plating thickness, and NiO contaminant film.

The remainder of this paper is organized as following: Section 2 presents the theory of some leading hypotheses on electrical contact resistance variation. Section 3 summarizes the modeling flow, assumptions and limitations. Section 4 provides the results and discussions, and Section 5 gives summary and conclusions.

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II. Some Hypotheses for Contact Resistance Variation

A. Contact Normal Force and Microhardness

In contact mechanics, it is well known that effective contact area is the sum of many asperities or A-Spots, and is directly related to contact normal force and material hardness. It can be calculated by using

$$A = \frac{F}{H} \tag{1}$$

where H is the Vickers microhardness and F is the contact normal force. The contact constriction resistance of bulk to bulk half space configuration can be expressed as the following equation reported by Holm [1]

$$R_c = \frac{\rho}{2a} = \frac{\rho}{2}\sqrt{\frac{\pi H}{F}} \tag{2}$$

where ρ is the resistivity of the contact material, a indicates the radius of effective contact area and $A = \pi a^2$. Therefore, uniform high contact normal force and low microhardness are the items that a socket/connector mechanical designer typically would look at first to achieve low electrical contact resistance. However, the contact normal force distribution is usually not uniform over the pin field due to system mechanical design limitations for large pin-count sockets. Therefore, normal force variation could be a significant contributor to electrical contact resistance variation. On the other hand, the microhardness of the plating system dependents on the plating composition and thickness. The change in microhardness also directly impacts the effective contact area at the interface. The bulk material properties of the given plating constituents are listed in TABLE I.

TABLE I. Bulk Material Properties of Plating Constituents

Material	Electrical Resistivity (μΩ-cm)	Vickers Microhardness (kg-mm^{-2})
Bulk Ni	8	300
Bulk Pd	10.8	500
Bulk Au	2.3	70

B. Phorsphorous in Electroless Nickel

One of the major differences between electroplated nickel and electroless plated nickel is purity. The purity of electroplated nickel is very high, commonly greater than 99%. For electroless nickel, the purity is typical 94% nickel and 6% phosphorus (P) because of the use of sodium hypophosphite reducing agent. The side-product phosphorous in electroless nickel has great effect on plating properties depending on its content percentage. 2-5% of P is typically considered as low level, 6-9% medium level and 10-13% high level [9]. For example, the microstructure of the electroless nickel is greatly different from electroplated nickel's crystalline structure. An amorphous nature becomes more dominant as the phosphorous content percentage increases. The amorphous structure is beneficial for base material diffusion prevention due to its lack of structured grain boundaries. On the other hand, other physical properties, such as hardness and electrical resistivity correspondingly change with P content percentage which could impact electrical contact resistance at the interface. Therefore, it is worth investigating the impact of P content percentage on the given multi-layer plating system.

The resistivity of bulk nickel is around 8 μΩ-cm. As shown in [9], the resistivity of electroless nickel can be 10 times greater with ~8% P due to impurity scattering of electrons.

P also has impact on microhardness. The microhardness of electroless plated nickel with P is higher than electroplated nickel. For example, the pure nickel Vickers hardness is around 300, versus 500 that of electroless nickel with ~8% P [9]. For electroless plated nickel with different P content percentages, the microhardness decreases as P content percentage increases [9].

Although electroless nickel is an under plating layer for CPU package pads, it could have main influence on interface contact resistance if its thickness is dominant. P content percentage in electroless nickel accordingly is critical to electrical contact resistance.

C. Nanofilm Size Effect

In "free electron theory", bulk resistivity seen in metal is due to scattering mechanisms of electrons moving in a crystal lattice, i.e. phonons (vibration of lattice atoms), impurities and point defects (vacancies). When the thickness of a metal film is reduced to the order of electron mean free path in nanometric scale, the "size effect" becomes dominant: the electron moving in thin metallic film will suffer from other two scattering mechanisms: Grain boundary (GB) scattering and surface scattering. Consequently, the thin metallic film electrical resistivity increases rapidly as thickness decreases. Various theories have been developed to study this phenomenon.

1) FS model- Surface scattering:

Fuchs and Sondheimer [4, 5] developed a theoretical model by considering the quantum effect of the free electrons with a distribution of the electron free path length λ values in the bulk. Film surface play an important role in the λ value is assumed.

$$\frac{\rho_0}{\rho_{FS}} = \frac{k}{\Phi_p(k)}$$

$$\frac{1}{\Phi_p(k)} = \frac{1}{k}\frac{3}{2k^2}(1-p)\int_1^\infty \left(\frac{1}{t^3} - \frac{1}{t^5}\right)\frac{1-e^{kt}}{1-pe^{-kt}}dt \tag{3}$$

where ρ_{FS} is the film resistivity value considering electron scattering at thin film surfaces , ρ_0 is the bulk resistivity value, $k = t/\lambda_0$ is the ratio of metallic film thickness t and λ_0 the corresponding bulk material mean free path, p is the fraction of elastically dispersed electrons by thin film surfaces: $p = 0$ for total surface scattering diffusely with max ρ_{FS}, while $p = 1$ for specular surface scattering and $\rho_{FS} = \rho_0$.

2) MS model- Grain boundary scattering:

When the grain size of thin metallic films is in the same order of electron mean free path, the electron scattering at these grain boundaries is high and leads to high resistity. Mayadas and Mayadas (MS) [6] proposed a model considering the dispersion of electrons by GB.

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$$\frac{\rho_0}{\rho_{MS}} = \left[1 - \frac{2}{3}\alpha + 3\alpha^2 - 3\alpha^3 ln\left(1 + \frac{1}{\alpha}\right)\right]^{-1}$$

$$\alpha = \frac{\lambda_0}{D}\left(\frac{R}{1-R}\right) \quad (4)$$

where ρ_{MS} is the film resistivity value considering electron scattering at GB, D is the grain size, R is the reflection coefficient ranging from 0 to 1 with $R = 0$ leading to max GB scattering, and when $R = 1$, electrons confined to individual grains and resistivity $\rho_{MS} = \rho_0$.

3) Combined model:

Juan and A.I. [7] proposed a combined theory of FS and MS models to account for both effects. By fitting the experimental measurements with combined equation, the corresponding p and R values for nanofilms are obtained. The combined model is obtained by superposition of Eqs. (3) and (4), and the total resistivity ρ_{total} can be expressed as

$$\rho_{total} = \rho_{FS} + \rho_{MS} - \rho_0 \quad (5)$$

Extensive research have been done to evaluate p and R value for different materials and film thicknesses. It is reported in [7] that for the Al, Au and Cu films by thermal evaporation with thickness from 3-100 nm on glass substrate, the surface scattering contribution is small comparing to the GB scattering contribution and both decrease with film thickness. In other word, p and R value drop with film thickness decreases. It is also shown that grain size (D) and its distribution is a function of thickness. The measured Al, Au, and Cu nanofilm thickness and corresponding D are fitted by a scaling law $D \propto t^B$, with B values reported 0.2±0.072, 0.36±0.052, and 0.75±0.073, respectively. As shown in [8], the grain size (GS) distribution for small GS makes great contribution to resistivity change in thin metallic film. For example, Cu thin film resistivity, with standard deviations of GS log normal distribution (σ=0.1, 0.2 and 0.3), are respectively 703.92 μΩ-cm, 3492.21 μΩ-cm and 8085.1 μΩ-cm at the same D=2nm. The resistivity with σ=0.1 is more than 10 times of that with σ=0.3 [8].

The p and R values, as well as their relationship with film thickness could change for different plating techniques and multi-layer plating configurations which are requires further investigation. However, this is beyond the scope of this study. In this paper, the behavior of metallic nanofilm thickness range described in [7] and [8] is used in modeling for general understanding of overall impact of the nanofilm "size effect" on electrical contact resistance of the given application. The assumed Au and Pd nanofilm electrical resistivity is listed in TABLE II.

D. NiO contaminant at the interface

Due to the existence of porosity, contaminants in the environment, high temperature exposure during manufacturing and Au plating wear during system cyclic assembly/ disassembly, exposed nickel is likely to form NiO contaminant at contact interface, and the contaminant film resistance can be written as

$$R_f = \frac{\rho_{cont}\, d}{A} = \frac{\rho_{cont}\, dH}{F} \quad (6)$$

where ρ_{cont} is the contaminant film material resistivity, and d is the film thickness. The electrical resistivity of pure and stoichiometric NiO is in the order of 10^{13} Ω-cm. Due to non-stoichiometry and surface state of NiO thin film, its resistivity is reported to be in the order of 10-10^6 Ω-cm depending on thickness [10, 11], which is 10^6-10^{11} times that of bulk nickel. As a result, it acts as an insulation layer at interface and has huge impact on electrical resistance at SLI interface. Hence, it is very important to control NiO growth at the SLI interface.

TABLE II. MATERIAL ELECTRICAL RESISTIVITY

Material	Electrical Resistivity (μΩ-cm)	Notes
Thin Pd film	1000-10000	~100-1000 times of bulk Pd
This Au film	250-2500	~100-1000 times of bulk Au
Thin film NiO	10^7-10^{12}	10^6-10^{11} times of bulk Ni

E. Surface Roughness, Porosity and Au Plaitng Thickness

With the 5-50gf contact normal force range in connector application, the asperity deformation is mainly in plastic deformation range [12]. Thus, surface roughness is a secondary factor for effective electrical contact area comparing to microhardness. On the other hand, from corrosion perspective, it is known that surface roughness is associated with Au plating thickness and porosity. This is due to plating thickness is typically related to plating uniformity which directly affects surface roughness and porosity. For example, in the experimental work of [13], the measurement shows that as Au plating thickness decreases, the surface roughness increases. Besides, porosity related corrosion leads to oxide formation, especially NiO. The porosity can be prevented by having uniform plating which is achievable through plating process optimization and Au plating thickness increase.

III. MULTI-LAYER PLATING SYSTEM ELECTRICAL CONTACT RESISTANCE MODELING FRAMEWORK

The proposed electrical contact resistance modeling flow for multi-layer plating systems is illustrated in Fig. 1. The modeling input parameters include total A-spot area, A-spot thickness, plating material resistivity, and plating layer dimensions. Once all the input information are obtained, a quasi-static 3D electromagnetic field simulation tool- ANSYS Q3D Extractor software is used to model multi-layer plating system and obtain its electrical resistance.

The NiO contaminant film is not included in the 3D model. The reasons are: 100% coverage of NiO on total A-spot area could result in open since the resistivity of NiO is 10^6-10^{11} times of bulk Ni. If certain area coverage is assumed, the unknown NiO area coverage and thickness pose modeling challenge in determining its effect on contact resistance via simulations.

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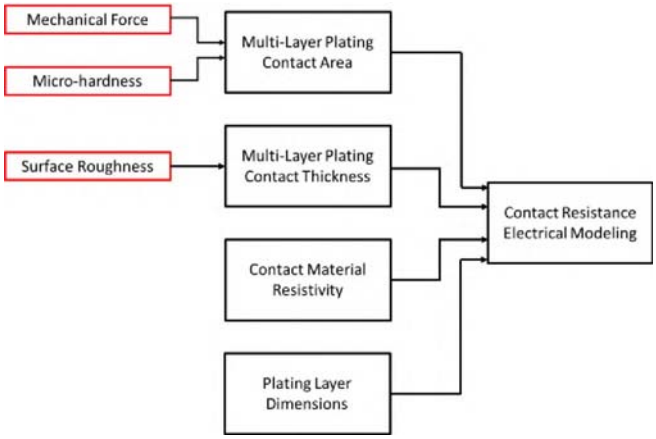


Fig. 1. Electrical contact resistance modeling flow.

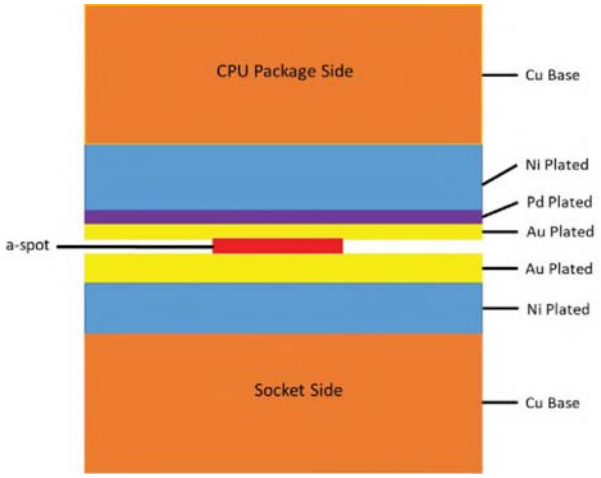


Fig. 2. Cross section of NiPdAu-plated CPU package pads and NiAu-plated second level interconnect socket contact interface.

The cross section of NiPdAu-plated CPU package pad and NiAu-plated SLI socket contact interface is shown in Fig. 2. For copper base electrical connections, on CPU package side, intermediate layers of electroless plated Ni and Pd are plated on copper based pad with a thin layer of immersion Au on top. On socket side, electroplated Ni and Au are plated on top of stamped copper alloy pin. Since the geometry of CPU package pad and socket pin is much greater than the effective contact area and has little impact on contact resistance, the geometries of both can be simplified as cylindrical conductors with radius b in EM simulation tool.

The total resistance of multi-layer plating is described as the sum of the contact constriction resistance R_c and the bulk resistance of plating layers R_{bulk}, illustrated in Eq. (7)

$$R_{model} = R_c + R_{bulk} \tag{7}$$

In general, due to the contact a-spot radius a << b, the contribution of these plating layers to the total electrical resistance is of little importance. Thus, the total resistance is mainly driven by constriction resistance.

Additionally, the A-spot thickness is assumed to be determined by surface roughness. It is found that with variation of A-spot thickness in the typical range of Au plating roughness reported in [13], the change in contact resistance is negligible. Therefore, it indicates that A-spot thickness variation in the given range is a secondary factor to contact resistance.

The proposed methodology has been firstly validated through comparison with Holm electrical contact resistance theoretical solution for bulk to bulk constriction resistance . In the following section, the effects including contact normal force, microhardness, P content percentage in electroless plated Ni, nanofilm size effect, and Au plating thickness, on the electrical contact resistance are numerically investigated using the proposed modeling framework.

IV. Results and Discussions

As shown in Eq. (1) and Eq. (2), the contact constriction resistance R_c at interface is directly related to A-spot radius dimension. The corresponding contact area depends on the contact normal force and micro-hardness. In this case, even a small change in A-spot radius may be translated to significant change in electrical contact resistance. Assuming constant microhardness, Fig. 3 shows that resistance increases exponentially as contact normal force decreases to less than 10gf. For contact normal force ranging from 12-50gf, the delta in resistance is about 2mOhm which is quite small. Therefore, a minimum contact normal force is required to achieve stable low electrical contact resistance. On the other hand, for the given case, increasing contact normal force beyond 20gf does not effectively reduce the resistance.

It is assumed that the microhardness drives the effective contact area at the interface will be dominated by the side with softer material. In the given case, the microhardness is mainly determined by the Au plating thickness on either CPU package side or socket side. It is observed in Fig. 4 that variation in Vickers microhardness from 50 to 500 (covering bulk Au, Pd and Ni microhardness), the resistance increases from 1.8mOhm to 6.5mOhm. This implies contact resistance is also sensitive to microhardness, while not as much as low contact normal force range. If electrical contact resistance needs to be tightly controlled at a low level, a low plating microhardness is required and can be achieved by having sufficient Au plating thickness.

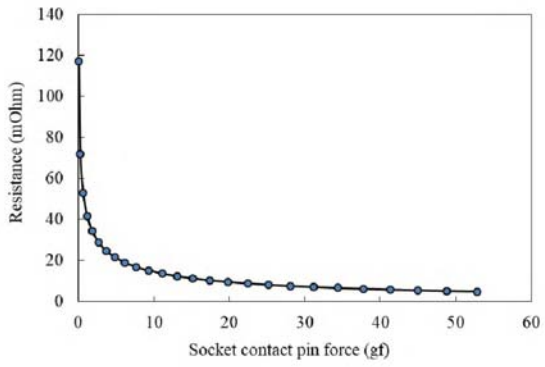


Fig. 3. Effect of socket contact contact normal force on eletrical contact resistance.

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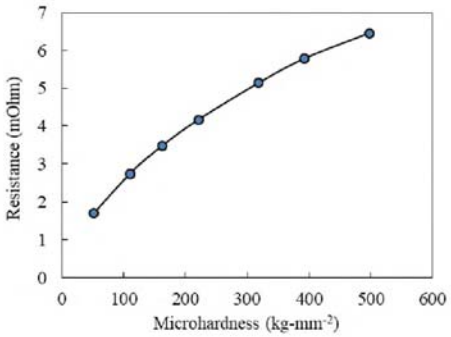


Fig. 4. Effect of microhardness on electrical contact resistance.

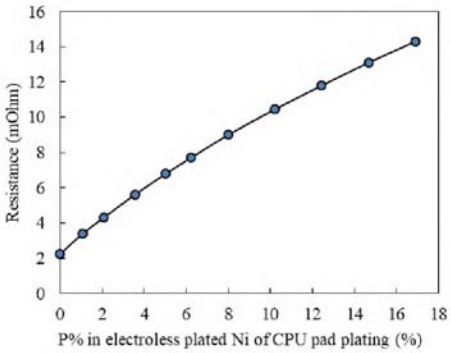


Fig. 5. Effect of P content percentage in electroless plated Ni of CPU package pad surface finish on electrial contact resistance.

Fig. 5 shows the dependence of electrical contact resistance on P content percentage in CPU package pad with electroless plated Ni. Even for low to medium P content percentage, i.e. 0-8%, the presence of phosphorus in electroless plated Ni layer increases the resistivity of Ni plating layer by up to 10 times [9]. Assuming the contact area remains constant, the resistance strongly hinges upon P content percentage. The resistance rises almost linearly with P content percentage, i.e. around 2mOhm for 0% and about 9mOhm for 8%. This is resulting from the dominant thickness of Ni in the given multi-layer plating on the CPU pad side, and its resistivity has major impact on the electrical contact resistance even it is an under plating layer.

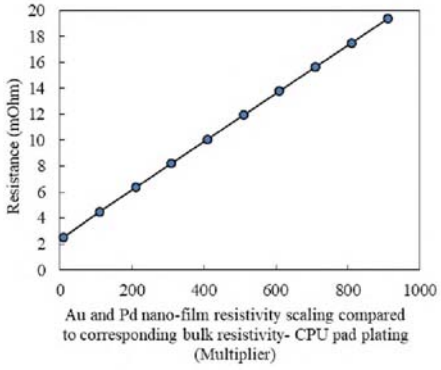


Fig. 6. Effect of Au and Pd nanofilm size effect on contact resistance.

As previously outlined in section II, for very thin metallic film, when the GS is at the same length scale of the bulk electron mean free path, the "size effect" would come into play on thin film resistivity, i.e. surface scattering and grain boundary scattering. The GS distribution variation contributes to additional change on thin film resistivity. As reported in [7] and [8], the resultant thin film resistivity can be as high as 1000 times of the corresponding bulk material resistivity. The size effect of CPU package pad side Au and Pd nanofilm on resistance is shown Fig. 6. If the thin film resistivity is 100 times of the bulk, the electrical contact resistance for the given multi-layer plating system is about twice of that assuming bulk resistivity. When 1000 times of bulk resistivity is assumed, the electrical contact resistance is about 10 times as much as that assuming bulk resistivity. Therefore, it is very important to maintain a minimum plating thickness so that the corresponding GS and GS distribution can be away from the "size effect" threshold. If very thin plating is needed for certain application, the relationship between plating thickness and GS/GS distribution, as well as their impact on electrical contact resistance should be well studied.

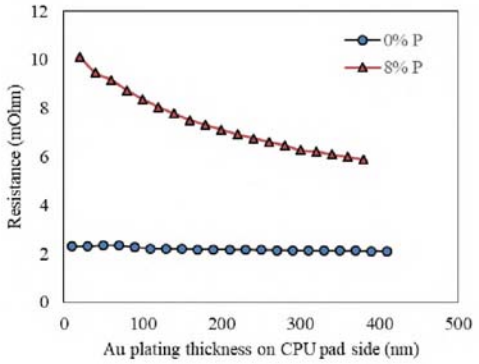


Fig. 7. Effect of Au plating thickness (CPU package pad side) on electrical contact resistance for two cases: 0% P and 8% P in electroless plated Ni.

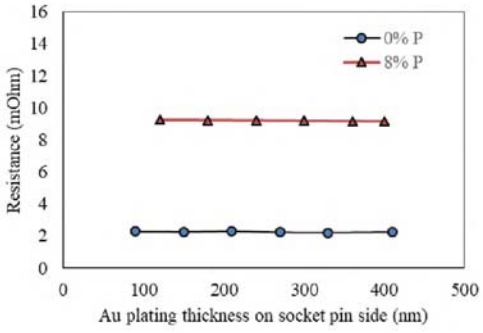


Fig. 8. Effect of Au plating thickness (socket pin side) on electrical contact resistance for two cases: 0% P and 8% P in electroless plated Ni.

Fig. 7 shows the effect of CPU pad side Au plating thickness on electrical contact resistance. Two cases with different P content percentage in electroless plated Ni have been studied. For CPU pad plating with 0% P, the electrical contact resistance for Au plating thickness ranging from 10-400nm is all around 2.2mOhm with negligible fluctuation. In contrast, for the case of

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8% P, the electrical contact resistance delta among different Au plating thicknesses is much greater. It is noticeable that contact resistance decreases as Au plating thickness increases. Therefore, it again demonstrates that electroless plated Ni layer has major impact on electrical contact resistance, as the resistance improvement due to Au plating thickness increase also strongly depends on it.

On the other hand, Fig. 8 shows the effect of socket contact side Au plating thickness on electrical contact resistance. Assuming the same total A-spot area, it can be seen that the resistance remains almost the same as Au thickness increases from ~100nm to 400nm. Therefore, the electrical contact resistance is insensitive to Au thickness itself in the given range on the socket contact side.

However, Au plating thickness at both CPU package and socket sides are very critical for other multi-fold reasons:

- Au is a corrosion barrier due to its nobility.
- Au is soft and can improve contact area.
- Au resistivity may be subject to thin film "size effect".
- Au thickness determines surface roughness and porosity.
- Electrical contact resistance becomes unstable when fretting corrosion occurs. There might be oxides and frictional polymer forming at the interface during fretting events, e.g. vibration. Palladium and its alloy are one of those materials very prone to fretting corrosion. The existence of Au can stabilize electrical contact resistance at the interface.

Finally, regarding NiO contaminant thin film which can be on either or both CPU package side and socket side, its resistivity could range 10^{7}-10^{12} μΩ-cm depending on the thickness and microstructure as reported in [11]. Assuming that NiO film cover area is 15μm in radius and 20nm in thickness, the calculated corresponding NiO contaminant film resistance is in the range of 3×10^{3}- 3×10^{8} mOhm. Clearly, NiO can be seen as an insulation layer as expected. Therefore, the effect of NiO on electrical contact resistance would mainly depend on its coverage of total A-spot area.

V. Summary and Conclusions

The electrical contact resistance of multi-layer plating system, i.e., NiPdAu plated CPU package pad and NiAu plated SLI socket contact has been numerically investigated. Multiple hypotheses on electrical contact resistance variation are quantified through modeling framework developed in this study. Based on the modeling results, the conclusions can be summarized as follows: First, resistance increases exponentially when contact normal force drops below a certain threshold while remains stable for high force. Decreasing the microhardness can improve the total A-spot area and reduce the resistance. The P content percentage in electroless plated Ni of CPU package pad surface finish is found to be another main driver for electrical contact resistance variation. As P content percentage increases, the microstructure of electroless plated Ni layer becomes more amorphous and results in a resistivity increase. Due to the dominant thickness of Ni in socket/connector application, its resistivity variation can be translate to change of electrical contact resistance. NiO contaminant film acts as an insulating layer once formed at the interface, and its impact is mainly determined by the NiO coverage of total A-spot area. Furthermore, for very thin metallic plating film with very small grain size, electrons experience scattering at surfaces and GBs, i.e. the so called "size effect" which could result in electrical contact resistance increase. Finally, Au plating thickness itself has minor impact compared to other factors mentioned above, but it is very critical to electrical contact resistance. This is for the reasons that Au could improve plating microhardness and uniformity, as well as reduce porosity. Existence of Au at contact interface can also prevent fretting corrosion under high frequency cyclic loading.

Based on the learning from modeling results, some potential design guidelines can be concluded from this study:

- Mechanical design optimization achieves uniform and relatively high contact normal force. But, above certain threshold, high contact normal force becomes less effective to reduce electrical contact resistance.
- P content percentage in electroless Ni sets the resistance level baseline for the given application and need to be well controlled.
- Adequate Au plating thickness can avoid "size effect", improve plating microhardness and uniformity (preventing porosity corrosion), as well as to mitigate fretting risk.
- Combining hard Au and soft Au plating could prevent wear during cyclic wiping and keep enough Au after wiping events to maintain a stable resistance at the interface.

The scope of this paper is limited to numerical investigation on the electrical contact resistance of given CPU package/socket SLI interface. To fully understand the attributes of electrical contact resistance for the given application, experimental validations are needed. Since there is not a lot of experimental work has been systematically done on CPU package/socket SLI electrical contact resistance, the next step for this study is to validate the findings from modeling results through experimental study.

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